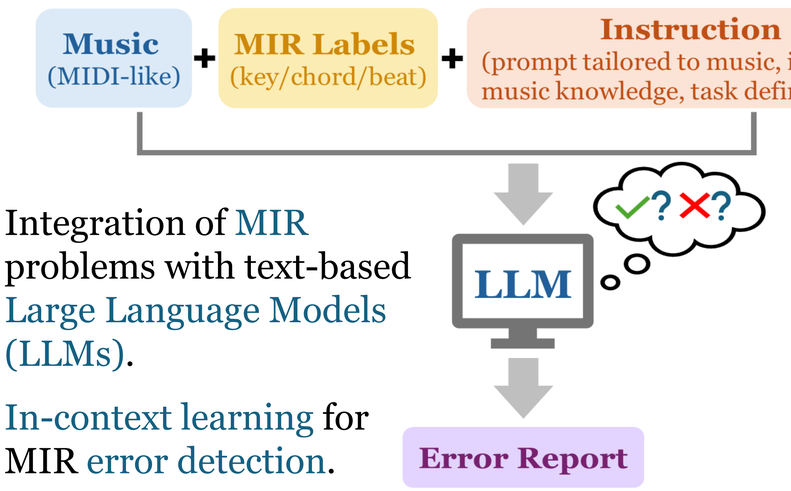
Exploring GPT's Ability as a Judge in Music Understanding

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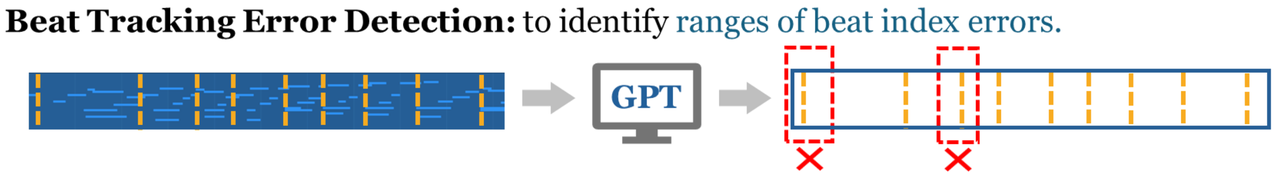
Introduction



- Integration of MIR problems with text-based Large Language Models (LLMs).
- In-context learning for MIR error detection.

Task Definition

Beat Tracking Error Detection: to identify ranges of beat index errors.



Chord Extraction Error Detection: to detect incorrect chord indices.



Key Estimation Error Detection: to determine correctness of key annotations.



Methodology

- **Prompt Structure**: (For chord analysis and beat tracking examples, please see the paper.)

Key Estimation Error Detection

Model Input

(Background & Task Definition)
Now I will give you a list of MIDI notes within several measures. An estimated key label for the music segment will be given. Please tell me whether the estimated key is wrong? Your final answer **should be "correct" or "incorrect".**

(Music Concepts Description)
In music, **key** refers to the central pitch or tonic note around which a piece of music revolves. It determines the tonal center and provides a framework for harmonies and melodies within a musical composition... (Some prompts are omitted here)

(Input Representation Explanation)
The input music and **key** estimation are as follows: key prediction: "C Major".
The measures you can use: {"measure_id": ..., "start": ..., "end": ..., "notes": [{"onset":..., "duration": ..., "pitch": ..., "velocity": ...}, ...]}
(Some prompts are omitted here)

(Techniques & Steps)
Here's a **step-by-step guide** to recognizing the key. Please think step by step to analyze the key prediction.
1. Calculate Pitch Distribution: ... **2.** Identify Dominant Pitch Classes: ... **3.** Determine Tonic Note: ... **4.** Analyze Context: ...
(Some prompts are omitted here)

(Output Format Specification)
The **final answer** should be in this format: [{"correctness": "correct" or "incorrect"}]

(Data Input)
Here is the given **MIDI notes and key prediction** for you to analyze: (Some prompts are omitted here)

Model Output

Analysis:
- The **key prediction** is Eb:min...
- Looking at the MIDI notes in the measures, we can observe the presence of pitches such as Eb (pitch 61), Gb (pitch 66), Bb (pitch 70), and Db (pitch 75). The notes align with the key prediction of Eb:min.

(Here omit some analysis by the model)
...
Based on the analysis above, the key estimation is: [{**"correctness": "incorrect"**}].

✗

+ Concepts

(Some prompts are omitted here)
1. ****Calculate Pitch Distribution****: **Count the occurrences of each pitch class (C, C#, D, etc.) and determine their relative frequencies**...
2. **Identify **Dominant Pitch Classes****: **Identify the pitch classes with the highest frequencies in the pitch distribution**...
(Some prompts are omitted here)

Analysis Steps:
1. **Calculate **Pitch Distribution****:
- Measure 1: C# (58), F (54), G# (42)...
2. **Identify **Dominant Pitch Classes****: **F# (20), D# (17), G (16)**
3. **Determine Tonic Note**: F# appears most frequently and consistently... (Here omit some analysis by the model)

✓

- Concepts

(Some prompts are omitted here)
Now I will give you **a list of data**. Also, I will provide you with **a label** for the entire data. Please tell me whether the **label prediction** is correct or incorrect based on **the context of the data**...
(Some prompts are omitted here)

The provided data consists of four rows, each with various features denoted as **'feature_0'**, **'feature_1'**, and **'feature_2'**... Based on the provided data, it appears that the **label prediction** "Eb:min" is incorrect. It seems to represent numerical measurements or observations of various features.
(Here omit some analysis by the model)

✓

Experiments

Concept Augmentation	CPR↑	EDR_P↑	EDR_N↑	WS↑
Basic Concepts	0.6681	0.3728	0.1794	0.5607
+ "Rhythm"	**0.8533**	0.1496	0.0968	**0.6520**
- "Beat Location"(Music Attribute Masking)	0.6898	0.3720	0.2008	0.5792
- "Beat Tracking"(Task Masking)	0.5998	0.4010	0.2862	0.5296
- "Music"(Domain Masking)	0.2418	**0.7657**	**0.7061**	0.3785
Random	0.513 ± 0.0586	0.4891 ± 0.0564	0.3238 ± 0.0608	0.4843 ± 0.0274

(a) Evaluation results on beat tracking error detection

Concept Augmentation	p↑	r↑	f↑
Basic Concepts	0.6345	0.6948	0.6207
+ "Chroma"	**0.6996**	**0.7174**	0.6290
- "Root"; "Quality"; "Inversion"(Music Attribute Masking)	0.6503	0.6992	0.6376
- "Chord Extraction"(Task Masking)	0.6497	0.6947	0.6362
- "Music"(Domain Masking)	0.6848	0.7144	**0.6480**
Random	0.5812 ± 0.0032	0.5003 ± 0.0034	0.5213 ± 0.0033

(b) Evaluation results on chord extraction error detection

Concept Augmentation	p↑	r↑	f↑
Basic Concepts	0.5789	**0.6513**	0.5965
+ "Scale"	0.5847	0.6169	**0.5972**
- "Tonic"; "Mode"(Music Attribute Masking)	0.5754	0.5812	0.5782
- "Key Estimation"(Task Masking)	0.5840	0.6143	0.5960
- "Music"(Domain Masking)	**0.5927**	0.4161	0.4085
Random	0.5779 ± 0.0086	0.4977 ± 0.0093	0.5186 ± 0.0089

(c) Evaluation results on key estimation error detection

Table: Evaluation results of GPT on three MIR error detection tasks: beat tracking, chord extraction, and key estimation. Each task is assessed under different concept augmentation strategies. "+" denotes Concept Introduction, and "–" denotes Concept Masking. " ↑ " indicates that higher values are better for all metrics. For beat tracking, Correct Pass Rate (CPR), Error Detection Rate on False Positives (EDR_P), Error Detection Rate on False Negatives (EDR_N), and the weighted score (WS) are reported. For chord extraction and key estimation tasks, precision (p), recall (r), and F-score (f) are reported.

Datasets

- **MIDI Datasets**
  - Classical piano (beat tracking).
  - Chinese Pop (chord and key).

	Beat Tracking	Chord Extraction	Key Estimation
#Notes	70,607	48,919	177,535
#Labels	14,194	9,200	2,271
#Tokens (per call)	6065.31	9256.80	3214.63

Conclusion

- GPT outperforms random baselines in beat tracking, chord extraction, and key estimation.
- Performance improves with better musical concepts in the prompts.
- Establishes a baseline for future LLM-based MIR research.

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